

Decision-time forward pass: legal actor path

Offline oracle path

Logged evidence

RGB, pose, calibration
semi-dense or fused
geometry
frozen EVL features /
support

Candidate rows

finite poses $\mathcal{Q}_t$
protocol-legal row
features
proposal provenance
explicit

Authoritative hard mask

$m_{t,i}^{\text{act}}$
gates selection, loss,
bootstrap
never changes raw Q_H

Reason codes explain rejected
rows; they are neither
embeddings nor low rewards.

Actor state

s_t^{cf0} , target descriptor φ_e
selected-view history
remaining budget

Q_H candidate scorer

f_θ on $(s_t, e, q_{t,i}, h)$
one row score per
materialized row

Prediction and action

raw $Q_{h,\theta,e,i}^{\text{cond}}$ per row
masked selection
downstream

Training / evaluation only

prediction–target
comparison
loss, metrics, named
ablations

Privileged scene assets

mesh, GT boxes, target
crop
identity and evaluation
support

Candidate renders

hard-valid candidate
depth / points
target endpoint error

Oracle targets

one-step target RRI
bounded returns and
upper bounds

gates loss

supervision